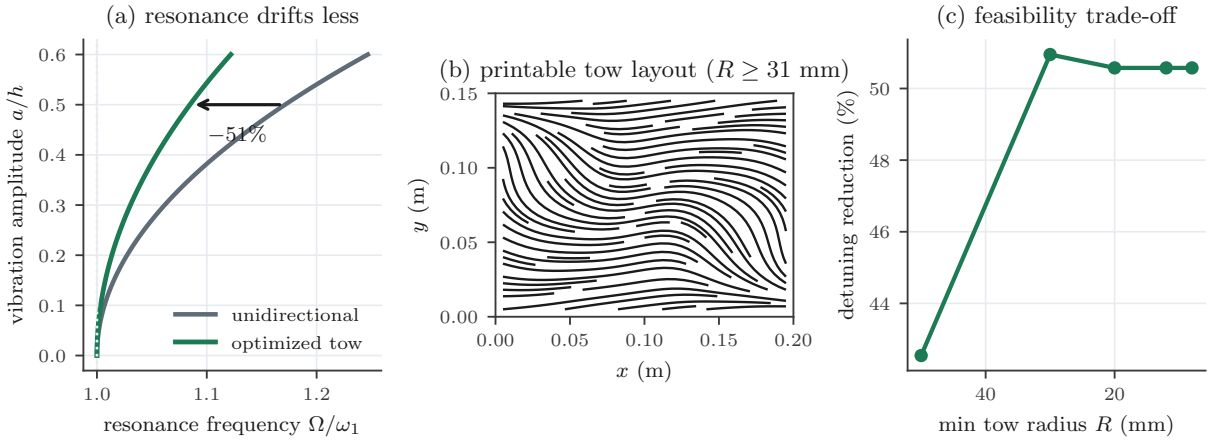

(a) resonance drifts less
vibration amplitude a/h
0.6
0.5
0.4
0.3
0.2
0.1
0.0
−51%
unidirectional
optimized tow
1.0
1.1
1.2
resonance frequency Ω/ω_1
(b) printable tow layout ($R \geq 31$ mm)
y (m)
0.15
0.10
0.05
0.00
0.00
0.05
0.10
0.15
0.20
x (m)
(c) feasibility trade-off
detuning reduction (%)
50
48
46
44
40
20
min tow radius R (mm)